## Revision Record

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# 1 Low-Level Protocol

TCP interface communication/port communication is based on byte stream, there is no message boundary. HL7 of high-level protocol is based on messages. The function of terminating the message is not provided. In order to determine the message boundary, the MLLP low-level protocol is used (*see HL7 Interface Standards Version 2.3* .1). Messages are transmitted in the following format:

<SB> *ddddd* <EB><CR>

among which:

**<SB> = Start Block character (1 byte)**

ASCII <VT>, namely, <0x0B>. Do not confuse with the SOH or STX character in ASCII.

***ddddd* = Data (variable number of bytes)**

*ddddd* is valid data of the HL7 information, and is displayed as UTF-8 code string.

**<EB> = End Block character (1 byte)**

ASCII <FS>, namely <0x1C>. Do not confuse with the ETX or EOT character in ASCII.

**<CR> = Carriage Return (1 byte)**

ASCII carriage return character, i.e.<0x0D>.

# 2 HL7 Message Constructing Principles

Every HL7 message consists of several segments and ends up with the <CR>.

Each segment consists of the segment name of three characters and a number of fields, and each field consists of some components and subcomponents. The first segment of each segment is MSH segment, which includes the field, unit and unit of delimiter.

MSH|^~\&| | |||1231235941||ORU^R01|2|P|2.3.1||||||UNICODE

Exampl MSH segments

The five characters following MSH segment define the delimiters used between fields, components and subcomponents. Although they can be any non-text characters, HL7 standard recommends you use the characters in the table below:

| Character | It means... |
|---|---|
| \| | Field delimiter |
| ^ | Component delimiter |
| & | Subcomponent delimiter |
| ~ | Repetition delimiter |
| \ | ESC |

The first field of MSH contains all the delimiters. Some fields behind are null because they are optional and not used by Mindray HL7 interface. Details about field definition and selection will be stated below.

For message of any type, the segments behind MSH appear in a fixed order. The order will be described in the following sections and the following grammar is used to organize the segments in proper order.

[] encloses optional segments.

{ } encloses segments which can repeat once or more.

# 3 Principles of Escape Character Conversion

For the field data of ST, TX, FT, and CF, etc. delimiters may be used in strings like remarks, clinical diagnosis and customized gender etc. When coding, the delimiters in the original strings shall be converted to escape sequence; which is restored in decoding. The principles for escape character conversion for HL7 interface are as follows:

| ESC Sequence | Original Character |
|---|---|
| \F\ | Field delimiter |
| \S\ | Component delimiter |
| \T\ | Subcomponent delimiter |
| \R\ | Repetition delimiter |
| \E\ | Escape delimiter |
| \.br\ | <CR>, segment end character. |

Note: the "\" in the escape sequence represents the ESC delimiter, whose value is defined in the MSH segment.

# 4 Communication

## 4.1 Process of communication

### 4.1.1 The main unit directly sends the test results

The PC connecting the analyzer actively sends the test results to the LIS. Test results and QC data are all send this way. The communication process is shown in Figure 1.

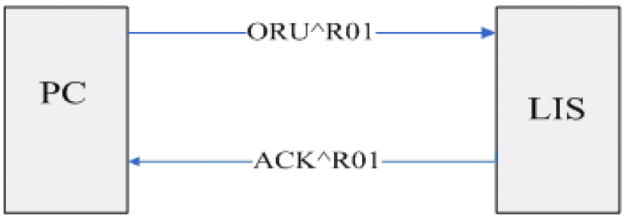


**Figure 1** Test results communication process

### 4.1.2 Worklist communication

Worklist belongs to the Order message. Thus, the corresponding HL7 messages: ORM(General Order Message) and ORR(General Order Response Message) can be used. The communication process is shown in Figure 2.

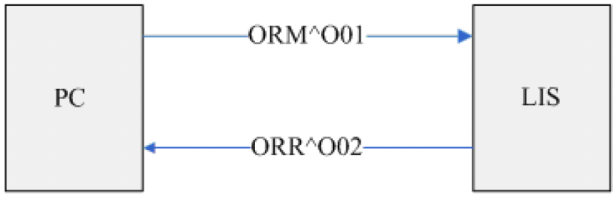


**Figure 2** Worklist searching communication process

## 4.2 Message Information

During the communication process, the message construction involves ORU^R01 message and ACK^R01 message. ORU^R01 message and ACK^R01 message appeared twinning, and is used for the communication of the analysis result and QC data.

### 4.2.1 ORU^R01 message

**ORU^R01 message**: is mainly used for the transmission of the analysis results and QC data.

ORU Observational Results (Unsolicited) Description

MSH Message Header, mandatory, including the communication information like message No., sending time, message delimiter and coding method, etc.

{

PID Patient demographic information, including patient name, gender, patient ID, date of birth, etc.

[PV1] Patient visit information, including patient type, department, bed No. and payer*, etc.

{

OBR sample information, including sample No., operator and time of analysis, etc.

{[OBX]} analysis data, including analysis results and mode of analysis, etc.

}

}

### 4.2.2 ACK^R01 message

**ACK^R01 message**: it confirms the receival of ORU^R01 message

ACK Acknowledgment Description

MSH Message head

MSA message acknowledgment, describing whether it has received the transmitted message

### 4.2.3 ORM^O01 message

**ORM^O01 message:** Common order message, all the actions related to order basically uses the message of this type. For example, create a new order or cancel an order. Here, the main unit requests LIS to re-fill the order message.

ORM General Order Message Description

MSH Message header

{ORC} Common message of Order, including the No. information of the sample searched

### 4.2.4 ORM^O02 message

**ORR^O02 message:** affirming of the ORM^O01 message. Here, returning the completed information of order (i.e. worklist).

ORR^O02 General Order Response Message Description

MSH Message header

MSA Message affirm

[

PID Patient basic information

[PV1] Patient visit information

]

{

ORC Common message of Order, including the sample No.

[

OBR sample information

{

[OBX] Data of other sample information, including work mode, etc.

}

]

}

## 4.3 Message Description

The tables in this section provide detailed definitions of the fields in all the message segments. Each row provides the information of one field, and the content of each column is described as follows:

1. No.: the HL7 message begins with the segment name of 3 characters followed by the fields which are separated by delimiters. "No." refers to the order of the field in the HL7 message segment.

E.g.

PID | 1 | |7393670^^^^MR||^Liu ||19950804000000|Female

↑ ↑ ↑

Segment name Field 1 Field 3

Note: for MSH segment, the field delimiter following the segment name is considered to be the first field, used to define the field delimiter values of the whole message.

2. Field name: the logic sense of the field.
3. Data type：the data type based on HL7 standards, See *6.1 HL7 Data Type Definition* for details;
4. Recommended max length: the recommended max length based on HL7 standards. But during the communication process, the data length may be longer than recommended, in which case the fields shall be identified by delimiters mark while analyzing the message segment.
5. Description: description to the value of the field.
6. Example: example of the fields.

### 4.3.1 MSH

MSH (Message Header) segment contains basic information of HL7 messages, including delimiter value, message type and coding method etc. It is the first field of every HL7 message.

Message example:

MSH|^~\&| |Mindray|||20210312092538||ORU^R01|2|P|2.3.1||||||UNICODE

For the value of each filed of the MSH message, see Table 4-1.

Table 4-1 Meaning table of the MSH message

| No. | Field/ delimiter Name | Data Type | Recommended Max Length | Description | Example |
|---|---|---|---|---|---|
| 1 | Field Separator | ST | 1 | Includes the delimiter of the first field after the segment name; used to | \| |

| | | | | | |
|---|---|---|---|---|---|
| | | | | determine the delimiter values of the rest part of the message. | |
| 2 | Encoding Characters | ST | 4 | Includes component delimiters, repetition delimiters, escape delimiters and subcomponent delimiters. | ^~\& |
| 3 | Sending application | EI | 180 | Application of sending terminal. | |
| 4 | Sending Facility | EI | 180 | Device of sending terminal. | Mindray |
| 7 | Date/Time Of Message | TS | 26 | Time of creating the message (in the format of YYYY[MM[DD[HH[MM[SS]]]]]), using the system time | 20210312092538 |
| 9 | Message Type | CM | 7 | Message type, in the format of "message type^event type". | ORU^R01 |
| 10 | Message Control ID | ST | 20 | Message control ID, used as the unique identifier of a message. | 2 |
| 11 | Processing ID | PT | 3 | Message processing ID. Value:<br>"P": sample and worklist searching message;<br>"Q": QC analysis result message;<br>In Ack messages, it is consistent with the previously received message. | P |
| 12 | Version ID | VID | 60 | HL7 version number. Value: "2.3.1". | 2.3.1 |
| 18 | Character Set | ID | 10 | Character set.<br>Value: "UNICODE", which means the message is expressed in UTF-8 strings. | UNICODE |

## 4.3.2 MSA

MSA（Message Acknowledgment）message includes the confirmation information, which appears in the Bi-Directional Response Message.

Message example:

MSA|AA|1

See Table 4-2 for field definitions in use.

Table 4-2 MSA Field Definitions

| No. | Field/ delimiter Name | Data Type | Recommended Max Length | Description | Example |
|---|---|---|---|---|---|
| 1 | Acknowledgment Code | ID | 2 | Acknowledgment code:"AA"-received; "AE" – error; "AR"-rejected. | AA |
| 2 | Message Control ID | ST | 20 | Message control ID, consistent with the MSH-10 of the received message | 1 |
| 6 | Error Condition | CE | 100 | Error condition (status code); it also contains error condition specification information; see Table 4-3 for the value. | |

Table 4-3 Error code of MSA-6 field

| Status code (MSA-6) | Status text (MSA-3) | Description/Remark |
|---|---|---|
| Successful: | | AA |
| 0 | Message accepted | Successful |
| Error status code: | | AE |
| 100 | Segment sequence error | Segment order in the message is wrong, or necessary segment lost |
| 101 | Required field missing | Necessary field lost in a segment |
| 102 | Data type error | Segment data type error, e.g. numbers are replaced by characters |
| 103 | Table value not found | Table value is not found; not used temporarily |
| Rejection status code: | | AR |
| 200 | Unsupported message type | Message type is not supported |
| 201 | Unsupported event code | Event code is not supported |
| 202 | Unsupported processing id | Processing ID is not supported |
| 203 | Unsupported version id | Version ID is not supported |
| 204 | Unknown key identifier | Unknown key identifier, e.g. transmitting the patient information that is not exited |

| | | |
|---|---|---|
| 205 | Duplicate key identifier | Repeated key words existed |
| 206 | Application record locked | Issues can not be executed at application program saving level, e.g. database is locked |
| 207 | Application internal error | Other interior errors of application program |

### 4.3.3 PID

The PID (Patient Identification) segment contains the patient demographic information.

Message example :

PID|1||7393670^^^^MR||Jerry^Tom||20210312092538|Male

See Table 4-4 for field definitions in use.

Table 4-4 Definition table of the PID field

| No. | Field/ delimiter Name | Data Type | Recommended Max Length | Description | Example |
|---|---|---|---|---|---|
| 1 | Set ID - PID | SI | 4 | Serial No., used to identify different PID segments in a message | 1 |
| 3 | Patient Identifier List | CX | 20 | Used as patient ID in the sample analysis result messages, in the form of "patient ID^^^^MR".<br>Used as batch No. of control in QC messages. | 7393670^^^^MR |
| 5 | Patient Name | XPN | 48 | Patient name (consists of Animal Name and Client), in the form of "Animal Name^Client" | Jerry^Tom |
| 7 | Date/Time of Birth | TS | 26 | Used as time of birth in sample information messages.<br>In the form of YYY[MM[DD[HH[MM[SS]]]]].<br>Used as expiration date of the control in QC messages. | 20210312092538 |
| 8 | Sex | IS | 1 | Gender, string. | Male |

### 4.3.4 PV1

The PV1 (Patient Visit) segment contains the patient visit information.

Message example:

PV1|1|Dog

See Table 4-5 for field definitions in use.

Table 4-5 PV1 Field Definitions

| No. | Field/ delimiter Name | Data Type | Recommended Max Length | Description | Example |
|---|---|---|---|---|---|
| 1 | Set ID - PV1 | SI | 4 | Serial No., used to identify different PV1 segments in a message. | 1 |
| 2 | Species | IS | 1 | Species, string, content not defined. | Dog |

## 4.3.5 OBR

The OBR (Observation Request) segment contains the test report information.

Message example:

OBR|1||TestSampleID1|00001^Automated Count^99MRC||20101006084439|20101009091515|||Li|||Cold|20101007084458|||||||||HM|||||||admin

See Table 4-6 for field definitions in use.

Table 4-6 OBR Field Definitions

| No. | Field/delimiter Name | Data Type | Recommended Max Length | Description | Example |
|---|---|---|---|---|---|
| 1 | Set ID - OBR | SI | 10 | Serial No., used to identify different OBR segments in a message | 1 |
| 2 | Placer Order Number | EI | 22 | Used as sample ID in the worklist searching response messages (i.e. ORR^O02 messages). | |
| 3 | Filler Order Number + | EI | 22 | Used as sample ID in sample analysis result messages.<br>Used as QC file No. in QC messages. | TestSampleID1 |
| 4 | Universal Service ID | CE | 200 | Universal service ID, used to identify different types of analysis results. See the enumeration constant sampling section for details. | 00001^Automated Count^99MRC |
| 6 | Requested Date/time | TS | 26 | Draw time.<br>Used as the time when the blood sample is drawn. | 20101006084439 |

| | | | | | |
|---|---|---|---|---|---|
| 7 | Observation Date/Time # | TS | 26 | Time of analysis. | 20101009091515 |
| 10 | Veterinarian | XCN | 60 | Veterinarian | Li |
| 13 | Relevant Clinical Info. | ST | 300 | Relevant clinical information.<br>Can be used as the clinical diagnostic information of patient information. | Cold |
| 14 | Specimen Received Date/Time * | TS | 26 | Time when the sample is received.<br>Used as the time when the analysis is ordered. | 20101007084458 |
| 15 | Specimen Source * | CM | 300 | Source of the sample.<br>Reserved field. | |
| 22 | Results Rpt/Status Chng - Date/Time + | TS | 26 | Result report/Status change - Tie.<br>Used as the time of validation. | |
| 24 | Diagnostic Serv Sect ID | ID | 10 | Diagnosis maker ID; value: "HM" (means Hematology) | HM |
| 28 | Result Copies To | XCN | 150 | Copy the result to.<br>Used as the person who validate the sample results. | |
| 32 | Principal Result Interpreter + | CM | 200 | Principal result interpreter.<br>Used as the operator of the sample analysis in sample messages.<br>Used as the operator of the QC count in QC messages. | admin |

## 4.3.6 OBX

The OBX (Observation/Result) segment contains the parameter information of each test result.

Message example:

OBX|7|NM|6690-2^WBC^LN||9.55|10*9/L|6.00-17.00||||F

See Table 4-7 for field definitions in use.

Table 4-7 OBX Field Definitions

| No. | Field/delimiter Name | Data Type | Recommended Max Length | Description | Example |
|---|---|---|---|---|---|
| 1 | Set ID - OBX | SI | 10 | Serial No., used to identify | 7 |

| | | | | | |
|---|---|---|---|---|---|
| | | | | different OBX segments in a message. | |
| 2 | Value Type | ID | 3 | Data type of the analysis result. Value: "ST", "NM", "ED", "IS", etc. | NM |
| 3 | Observation Identifier | CE | 590 | Analysis item identifier.<br>In the form of "ID^Name^EncodeSys", where ID is the identifier of the analysis item; Name is the description of the item; EncodeSys is the coding system of the item.<br>See the enumeration constant sampling section for details.<br>Note: ID and EncodeSys are used to identify different analysis parameters, while Name is for description purpose rather than identification. | 6690-2^WBC ^LN |
| 5 | Observation Value | * | 65535 | Analysis result data, which can be numeric, string, enumeration value, binary data, etc. (Binary data like histogram or scattergram are converted to codes using the Base64 coding method. See the following sections for the coding method). | 9.55 |
| 6 | Units | CE | 90 | Unit of analysis items. It adopts the ISO standard unit. See the enumeration constant sampling section for details. | 10*9/L |
| 7 | References Range | ST | 90 | Reference range of analysis results, in the form of "lower limit-higher limit", "<upper limit" or ">lower limit". | 6.00-17.00 |
| 8 | Abnormal Flags | ID | 5 | Analysis result flags. Value definitions: | |

| | | | | | |
|---|---|---|---|---|---|
| | | | | “N”- normal<br>“A”- abnormal<br>"H": higher than upper limit<br>“L” – lower than lower limit<br>Note: The flag for normal or abnormal and that for high or low result may appear in this field at the same time. In this case, the two types of flags are connected by a “~”, e.g. “H~A” | |
| 11 | Observ Result Status | ID | 1 | Status of the analysis result. "F" - （Final Result）, which refers to Final Result. | F |
| 13 | User Defined Access Checks | ST | 20 | User-defined. For flags of reagent expiration or modification, etc. In the form of”Flag1~Flag2”.<br>There are 3 types of flags in all:<br>O – reagent expiration<br>E – result edited flag<br>e – result changed due to the manual editing of another parameter result based on which it is calculated | |

### 4.3.7 ORC

The ORC(Common Order) segment contains the common information of order.

Message example:

ORC|RF||SampleID||IP

See Table 4-8 for definition of the fields used.

Table 4-8 ORC Field Definitions

| No. | Field/ delimiter Name | Data Type | Recommended Max Length | Description | Example |
|---|---|---|---|---|---|
| 1 | Order Control | ID | 2 | Order control word In the ORM message the value is “RF” which means “re-fill the order request”. In the ORR message | RF |

| | | | | | |
|---|---|---|---|---|---|
| | | | | the value is “AF” which means “affirm the re-filled order”. | |
| 2 | Placer Order Number | EI | 22 | Placer order number In the ORM message the value is empty; in the ORR message the value is the sample ID. | |
| 3 | Filler Order Num | EI | 22 | Filler Order Number In the ORM message the value is the sample ID; in the ORR message the value is empty. The value “*” means query work list. | SampleID |
| 5 | Order Status | ID | 2 | Order status In the ORM message the value is “IP” which means “order is being processed, but results are not obtained”; in the ORR message the value is empty. | IP |

# 5 Example

## 5.1 Worklist Query Message

MSH|^~\&|||||20150318142306||ORM^O01|5|P|2.3.1||||||UNICODE

ORC|RF||*||IP

## 5.2 Worklist Response Message

MSH|^~\&|LIS|LISSimulator|||20150109154446||ORR^O02|0|P|2.3.1||||||UNICODE

MSA|AR|5||||

PID|1||NO-4B00003^^^^^MR||Bob^Jerry||20150105|male

PV1|1|Species

ORC|AF|AT-JK00927

OBR|1||AT-JK00927|00001^Automated

Count^99MRC||20150318141722||||Bill|||_d_|20150318141722|||||||||HM

OBX|1|NM|30525-0^Age^LN||2|mo|||||F

OBX|2|ST|01001^Remark^99MRC||||||||F

OBX|3|IS|08001^Take Mode^99MRC||O||||||F

OBX|4|IS|08002^Blood Mode^99MRC||P||||||F

OBX|5|IS|08003^Test Mode^99MRC||CBC+DIFF||||||F

OBX|6|IS|01002^Ref Group^99MRC||0||||||F

OBX|7|IS|01003^Species^99MRC||724||||||F

OBX|8|XPN|01004^Master Name^99MRC||Bob^Jerry|||||F

## 5.3 Sample Message

MSH|^~\&|BC-60R Vet|Mindray|||20220820093119||ORU^R01|1|P|2.3.1||||||UNICODE

PID|1||^^^^MR|||||U

PV1|1|Dog

OBR|1||DE4-5-dog-CDR-2|00001^Automated Count^99MRC|||20220719151838||||||||||||||HM|||||||rd

OBX|1|IS|08001^Take Mode^99MRC||O||||||F

OBX|2|IS|08002^Blood Mode^99MRC||W||||||F

OBX|3|IS|08003^Test Mode^99MRC||CBC+DIFF+RET||||||F

OBX|4|IS|01002^Ref Group^99MRC||0||||||F

OBX|5|IS|05007^Project Type^99MRC||BL||||||F

OBX|6|ST|09003^SN^99MRC||YJ-DE041705||||||F

OBX|7|IS|01003^Species^99MRC||720||||||F

OBX|8|NM|6690-2^WBC^LN||8.77|10*9/L|5.32-16.92|N|||F

OBX|9|NM|704-7^BAS#^LN||0.01|10*9/L|0.00-0.13|N|||F

OBX|10|NM|706-2^BAS%^LN||0.1|%|0.0-1.0|N|||F

OBX|11|NM|751-8^NEU#^LN||4.62|10*9/L|3.05-12.10|N|||F

OBX|12|NM|770-8^NEU%^LN||52.7|%|42.0-84.0|N|||F

OBX|13|NM|711-2^EOS#^LN||0.52|10*9/L|0.04-1.28|N|||F
OBX|14|NM|713-8^EOS%^LN||5.9|%|0.3-11.0|N|||F
OBX|15|NM|731-0^LYM#^LN||3.13|10*9/L|0.70-4.95|N|||F
OBX|16|NM|736-9^LYM%^LN||35.7|%|6.0-40.0|N|||F
OBX|17|NM|742-7^MON#^LN||0.49|10*9/L|0.20-1.38|N|||F
OBX|18|NM|5905-5^MON%^LN||5.6|%|2.5-12.0|N|||F
OBX|19|NM|789-8^RBC^LN||6.05|10*12/L|4.70-8.69|N|||F
OBX|20|NM|718-7^HGB^LN||152|g/L|115-201|N|||F
OBX|21|NM|787-2^MCV^LN||74.8|fL|60.0-77.5|N|||F
OBX|22|NM|785-6^MCH^LN||25.1|pg|20.0-27.0|N|||F
OBX|23|NM|786-4^MCHC^LN||335|g/L|300-380|N|||F
OBX|24|NM|788-0^RDW-CV^LN||13.4|%|11.3-18.9|N|||F
OBX|25|NM|21000-5^RDW-SD^LN||38.2|fL|29.1-55.1|N|||F
OBX|26|NM|4544-3^HCT^LN||45.3|%|35.0-60.0|N|||F
OBX|27|NM|777-3^PLT^LN||232|10*9/L|140-520|N|||F
OBX|28|NM|32623-1^MPV^LN||12.2|fL|7.6-16.1|N|||F
OBX|29|NM|32207-3^PDW^LN||15.8||13.8-17.9|N|||F
OBX|30|NM|10002^PCT^99MRC||0.283|%|0.120-0.700|N|||F
OBX|31|NM|14196-0^RET#^LN||40.4|10*9/L|9.0-115.0|N|||F
OBX|32|NM|4679-7^RET%^LN||0.67|%|0.16-1.95|N|||F
OBX|33|NM|10015^LFR^99MRC||73.5|%|56.0-100.0|N|||F
OBX|34|NM|10016^MFR^99MRC||21.1|%|0.0-26.0|N|||F
OBX|35|NM|10017^HFR^99MRC||5.4|%|0.0-22.0|N|||F
OBX|36|NM|10022^PLT-I^99MRC||161|10*9/L||N|||F
OBX|37|NM|10024^WBC-D^99MRC||8.77|10*9/L||N|||F
OBX|38|NM|10018^RBC-O^99MRC||6.06|10*12/L||N|||F
OBX|39|NM|10019^PLT-O^99MRC||232|10*9/L||N|||F
OBX|40|NM|10013^PLCC^99MRC||78|10*9/L|25-180|N|||F
OBX|41|NM|10014^PLCR^99MRC||33.7|%|10.0-57.0|N|||F
OBX|42|NM|10041^IPF^99MRC||2.5|%|0.4-17.1|N|||F
OBX|43|NM|10043^RHE^99MRC||26.4|pg|20.0-28.3|N|||F
OBX|44|NM|33516-6^IRF^LN||26.5|%|0.0-45.1|N|||F
OBX|45|NM|48706-6^MRV^LN||93.5|fL||N|||F
OBX|46|NM|10031^PDW-SD^99MRC||15.9|fL||N|||F
OBX|47|NM|10051^WBC-O^99MRC||8.15|10*9/L||N|||F
OBX|48|NM|10060^RPI^99MRC||0.7|||N|||F
OBX|49|NM|10062^IPF#^99MRC||6|10*9/L||N|||F
OBX|50|IS|7796-6^Platelet Clump?^LN||T||||||F
OBX|56|NM|15051^RBC Histogram. Left Line^99MRC||0||||||F
OBX|57|NM|15052^RBC Histogram. Right Line^99MRC||0||||||F

OBX|58|NM|15053^RBC Histogram. Binary Meta Length^99MRC||1||||||F

OBX|59|NM|15057^RBC Histogram. Total^99MRC||0||||||F

OBX|60|ED|15056^RBC Histogram. BMP^99MRC|| ^Image^BMP^Base64^…||||||F

OBX|61|NM|15111^PLT Histogram. Left Line^99MRC||0||||||F

OBX|62|NM|15112^PLT Histogram. Right Line^99MRC||0||||||F

OBX|63|NM|15113^PLT Histogram. Binary Meta Length^99MRC||1||||||F

OBX|64|NM|15117^PLT Histogram. Total^99MRC||0||||||F

OBX|65|ED|15116^PLT Histogram. BMP^99MRC||^Image^BMP^Base64^…||||||F

OBX|71|NM|15203^WBC DIFF Scattergram. Meta len^99MRC||1||||||F

OBX|72|NM|15205^WBC DIFF Scattergram. Fsc dimension^99MRC||0||||||F

OBX|73|NM|15206^WBC DIFF Scattergram. Ssc dimension^99MRC||0||||||F

OBX|74|ED|15200^WBC DIFF Scattergram. BMP^99MRC||^Image^BMP^Base64^…||||||F

OBX|75|NM|15307^RET Scattergram. Meta Len^99MRC||1||||||F

OBX|76|NM|15303^RET Scattergram. Fsc dimension^99MRC||0||||||F

OBX|77|NM|15304^RET Scattergram. Ssc dimension^99MRC||0||||||F

OBX|78|ED|15302^RET-EXT Scattergram. BMP^99MRC||^Image^BMP^Base64…||||||F

OBX|79|ED|15301^PLT-O Scattergram. BMP^99MRC||^Image^BMP^Base64^…||||||F

OBX|80|ED|15300^RET Scattergram. BMP^99MRC||^Image^BMP^Base64^…||||||F

## 5.4 Sample Response Message

A sample response message needs to be responded after receiving a sample result each time. which contains two segments: MSH and MSA. To send a correct response message, take into consideration that: the MSH-9 field should be ACK^R01 which indicates that it is a sample response message; If the value in the MSA-2 field is the same with the MSH-10 value of the received analysis result, it indicates that this response message is corresponding to the sent analysis result. The MSA-2 value in the following example is 2.

MSH|^~\&|LIS||||20361231235956||ACK^R01|1|P|2.3.1|||||UNICODE MSA|AA|2

## 5.5 QC Message

The content form of QC Message is different from that of the sample count result message, the MSH-11 of the QC message is valued as Q, the represented message type is QC data.

BC-6D Normal Control

MSH|^~\&|BC-60R Vet|Mindray|||20220820093217||ORU^R01|11|Q|2.3.1|||||UNICODE

PID|1||MB0722AN||||20220910000000

OBR|1||2|00003^LJ QCR^99MRC|||20220815095140||||||||||||||||HM|||||||rd

OBX|1|IS|05001^Qc Level^99MRC||M||||||F

OBX|2|IS|08001^Take Mode^99MRC||O||||||F

OBX|3|IS|08002^Blood Mode^99MRC||W|||||F
OBX|4|IS|08003^Test Mode^99MRC||CBC+DIFF|||||F
OBX|5|NM|6690-2^WBC^LN||8.59|10*9/L|6.66-8.66|N|||F
OBX|6|NM|704-7^BAS#^LN||0.07|10*9/L|0.00-0.18|N|||F
OBX|7|NM|706-2^BAS%^LN||0.9|%|0.0-2.0|N|||F
OBX|8|NM|751-8^NEU#^LN||4.13|10*9/L|2.94-4.54|N|||F
OBX|9|NM|770-8^NEU%^LN||48.0|%|38.8-58.8|N|||F
OBX|10|NM|711-2^EOS#^LN||1.00|10*9/L|0.07-1.15|N|||F
OBX|11|NM|713-8^EOS%^LN||11.6|%|0.9-14.9|N|||F
OBX|12|NM|731-0^LYM#^LN||2.84|10*9/L|2.08-3.68|N|||F
OBX|13|NM|736-9^LYM%^LN||33.1|%|27.6-47.6|N|||F
OBX|14|NM|742-7^MON#^LN||0.55|10*9/L|0.00-0.81|N|||F
OBX|15|NM|5905-5^MON%^LN||6.4|%|0.0-10.6|N|||F
OBX|16|NM|789-8^RBC^LN||4.67|10*12/L|4.42-4.98|N|||F
OBX|17|NM|718-7^HGB^LN||136|g/L|128-142|N|||F
OBX|18|NM|787-2^MCV^LN||94.6|fL|87.6-99.6|N|||F
OBX|19|NM|785-6^MCH^LN||29.1|pg|25.7-31.7|N|||F
OBX|20|NM|786-4^MCHC^LN||307|g/L|272-342|N|||F
OBX|21|NM|788-0^RDW-CV^LN||15.0|%|11.0-21.0|N|||F
OBX|22|NM|21000-5^RDW-SD^LN||53.8|fL|44.4-64.4|N|||F
OBX|23|NM|4544-3^HCT^LN||44.1|%|41.0-47.0|N|||F
OBX|24|NM|777-3^PLT^LN||205|10*9/L|150-250|N|||F
OBX|25|NM|32623-1^MPV^LN||13.3|fL|10.8-16.8|N|||F
OBX|26|NM|32207-3^PDW^LN||17.9| |12.2-22.2|N|||F
OBX|27|NM|10002^PCT^99MRC||0.273|%|0.176-0.376|N|||F
OBX|28|NM|10013^PLCC^99MRC||102|10*9/L|82-130|N|||F
OBX|29|NM|10014^PLCR^99MRC||49.9|%|37.9-67.9|N|||F

BC-RET Normal Control

MSH|^~\&|BC-60R Vet|Mindray|||20220820093926||ORU^R01|116|Q|2.3.1|||||UNICODE
PID|1||ME0722A-2||||20220910000000
OBR|1||5|00003^LJ QCR^99MRC|||20220815195345||||||||||||||HM|||||||rd
OBX|1|IS|05001^Qc Level^99MRC||M|||||F
OBX|2|IS|08001^Take Mode^99MRC||O|||||F
OBX|3|IS|08002^Blood Mode^99MRC||W|||||F
OBX|4|IS|08003^Test Mode^99MRC||CBC+DIFF+RET|||||F
OBX|5|NM|6690-2^WBC^LN||1.33|10*9/L||N|||F
OBX|6|NM|704-7^BAS#^LN||0.01|10*9/L||N|||F
OBX|7|NM|706-2^BAS%^LN||0.5|%||N|||F

OBX|8|NM|751-8^NEU#^LN||1.14|10*9/L||N|||F
OBX|9|NM|770-8^NEU%^LN||85.7|%||N|||F
OBX|10|NM|711-2^EOS#^LN||0.18|10*9/L||N|||F
OBX|11|NM|713-8^EOS%^LN||13.5|%||N|||F
OBX|12|NM|731-0^LYM#^LN||0.00|10*9/L||N|||F
OBX|13|NM|736-9^LYM%^LN||0.3|%||N|||F
OBX|14|NM|742-7^MON#^LN||0.00|10*9/L||N|||F
OBX|15|NM|5905-5^MON%^LN||0.0|%||N|||F
OBX|16|NM|789-8^RBC^LN||4.54|10*12/L||N|||F
OBX|17|NM|718-7^HGB^LN||134|g/L||N|||F
OBX|18|NM|787-2^MCV^LN||95.3|fL||N|||F
OBX|19|NM|785-6^MCH^LN||29.6|pg||N|||F
OBX|20|NM|786-4^MCHC^LN||310|g/L||N|||F
OBX|21|NM|788-0^RDW-CV^LN||14.9|%||N|||F
OBX|22|NM|21000-5^RDW-SD^LN||53.8|fL||N|||F
OBX|23|NM|4544-3^HCT^LN||43.2|%||N|||F
OBX|24|NM|777-3^PLT^LN||253|10*9/L||N|||F
OBX|25|NM|32623-1^MPV^LN||9.3|fL||N|||F
OBX|26|NM|32207-3^PDW^LN||17.7| ||N|||F
OBX|27|NM|10002^PCT^99MRC||0.236|%||N|||F
OBX|28|NM|14196-0^RET#^LN||192.1|10*9/L|106.6-304.6|N|||F
OBX|29|NM|4679-7^RET%^LN||4.23|%|2.60-6.60|N|||F
OBX|30|NM|10015^LFR^99MRC||78.4|%|53.0-83.0|N|||F
OBX|31|NM|10016^MFR^99MRC||18.7|%|16.3-36.3|N|||F
OBX|32|NM|10017^HFR^99MRC||2.9|%|0.0-11.4|N|||F
OBX|33|NM|10013^PLCC^99MRC||59|10*9/L||N|||F
OBX|34|NM|10014^PLCR^99MRC||23.3|%||N|||F
OBX|35|NM|10041^IPF^99MRC||5.6|%|0.0-20.5|N|||F
OBX|36|NM|10043^RHE^99MRC||29.6|pg|17.1-41.1|N|||F
OBX|37|NM|33516-6^IRF^LN||21.6|%|19.0-45.0|N|||F

## 5.6 QC Response Message

The only difference between the QC response message and the sample analysis result response message is that the MSH-11 value of the QC response message is Q.

MSH|^~\&|LIS||||20110531120835||ACK^R01|1|Q|2.3.1||||||UNICODE
MSA|AA|2

## 5.7 Bidirectional LIS inquiry response message

When the LIS received an inquiry message, it needs to send back an inquiry response message. The first two message segments of the inquiry response message are MSH and MSA. The MSH-9 field (indicating the type of the segment) is filled with ORR^O02, while the MSA segment should be filled up as shown in the following example of the inquiry response message. If the LIS gets searching results for the inquiry, there will be PID, PV1, ORC, OBR and OBX message segments after the two heading segments to provide the patient and sample information, in the same way as the sample data message does. The ORC segment is indispensable for an inquiry response message with searching results, in which the ORC-1 value is AF, and ORC-2 is the filter (the sample ID). Note that the OBR-2 field indicates the sample ID, which should be the same value as in the ORC-2 field; Otherwise, the message will be regarded as incorrect.

An example of the inquiry response message with searching results is shown as follows:

MSH|^~\&|LIS||||20140709094045||ORR^O02|1|P|2.3.1||||||UNICODE

MSA|AA|4

PID|1||ChartNo^^^^^MR||^FName||19810506|Female

PV1|1||||||||||||||||||NewCharge

ORC|AF|SampleID1|||

OBR|1|SampleID1||||20060506||||tester|||Diagnose content....|20060504||||||||20080821||HM||||||||veterinarian

OBX|1|IS|08001^Take Mode^99MRC||A||||||F

OBX|2|IS|08002^Blood Mode^99MRC||W||||||F

OBX|3|IS|08003^Test Mode^99MRC||CBC||||||F

OBX|4|IS|01002^Ref Group^99MRC||XXXX||||||F

OBX|5|NM|30525-0^Age^LN||1|hr|||||F

OBX|6|ST|01001^Remark^99MRC||remark content....||||||F

An example of the inquiry response message with no search result is shown as follows, in which the MSA-2 field indicates the result of the response. In this example, the MSA-2 value is “AR”, indicating the inquiry was rejected; if it is “AE", then there is an error in the inquiry process.

MSH|^~\&|LIS||||20081120175238||ORR^O02|1|P|2.3.1||||||UNICODE

MSA|AR|1

## 5.8 HL7 Coding and Constant Definition

## 5.9 OBR-4 Message Type Coding

OBR-4 field is used to mark the analysis type, see Table 5-1.

Table 5-1 OBR-4 message type value table

| **Data** | **Code（ID）** | **Name** | **EncodeSys** |
|---|---|---|---|
| Sample Analysis Result | 00001 | Automated Count | 99MRC |

| | | | |
|---|---|---|---|
| Microscopic result | 00002 | Manual Count | 99MRC |
| LJ C count result | 00003 | LJ QCR | 99MRC |

## 5.10 OBX-3 parameter type code

Each OBX message field contains an analysis item or information of other data. OBX-3（Observation Identifier） is the mark of the data item, which displays as “ID^Name^EncodeSys”；OBX-5 contains the value of the data item; OBX-6 contains the parameter unit, which is displayed as the ISO standard unit.

Table 5-2 OBR-3 parameter type code table

| **Data** | **HL7 Type (OBX-2)** | **Code(ID)** | **Name** | **EncodeSys** | **Example of OBX-3 field** |
|---|---|---|---|---|---|
| Non-parameter Data Items | | | | | |
| Presentation mode | IS | 08001 | Take Mode | 99MRC | 08001^Take Mode^99MRC |
| Blood Mode | IS | 08002 | Blood Mode | 99MRC | 08002^Blood Mode^99MRC |
| Measurement Mode | IS | 08003 | Test Mode | 99MRC | 08003^Test Mode^99MRC |
| Age | NM | 30525-0 | Age | LN | 30525-0^Age^LN |
| Remarks | ST | 01001 | Remark | 99MRC | 01001^Remark^99MRC |
| Reference group | IS | 01002 | Ref Group | 99MRC | 01002^Ref Group^99MRC |
| Species | IS | 01003 | Species | 99MRC | 01003^Species^99MRC |
| Recheck flag | IS | 01006 | 99MRC | 01006^Recheck flag^99MRC | Recheck flag |
| Sample Type | IS | 01007 | 99MRC | 01007^Sample Type^99MRC | Sample Type |
| Shelf No | ST | 01012 | 99MRC | 01012^Shelf No^99MRC | Shelf No |
| Tube No | ST | 01013 | 99MRC | 01013^Tube No^99MRC | Tube No |
| Report Time | ST | 01014 | 99MRC | 01014^Report Time^99MRC | Report Time |
| Level of control | IS | 05001 | Qc Level | 99MRC | 05001^Qc Level^99MRC |
| Qc valid date | ST | 05004 | 99MRC | 05004^Qc valid date ^99MRC | Qc valid date |
| Qc file No | ST | 05005 | 99MRC | 05005^Qc file No ^99MRC | Qc file No |
| Qc lot No | ST | 05006 | 99MRC | 05006^Qc lot No ^99MRC | Qc lot No |
| Project type | ST | 05007 | 99MRC | 05007^Project Type^99MRC | Project type |
| Analyzer | ST | 09001 | 99MRC | 09001^Analyz | Analyzer |

| | | | | | |
|---|---|---|---|---|---|
| | | | | er^99MRC | |
| SN | ST | 09003 | 99MRC | 09003^ SN ^99MRC | SN |
| Parameter Result Items | | | | | |
| WBC | NM | 6690-2 | WBC | LN | 6690-2^WBC^LN |
| BAS | NM | 704-7 | BAS# | LN | 704-7^BAS#^LN |
| BAS_PER | NM | 706-2 | BAS% | LN | 706-2^BAS%^LN |
| NEU | NM | 751-8 | NEU# | LN | 751-8^NEU#^LN |
| NEU_PER | NM | 770-8 | NEU% | LN | 770-8^NEU%^LN |
| EOS | NM | 711-2 | EOS# | LN | 711-2^EOS#^LN |
| EOS_PER | NM | 713-8 | EOS% | LN | 713-8^EOS%^LN |
| LYM | NM | 731-0 | LYM# | LN | 731-0^LYM#^LN |
| LYM_PER | NM | 736-9 | LYM% | LN | 736-9^LYM%^LN |
| MON | NM | 742-7 | MON# | LN | 742-7^MON#^LN |
| MON_PER | NM | 5905-5 | MON% | LN | 5905-5^MON%^LN |
| RBC | NM | 789-8 | RBC | LN | 789-8^RBC^LN |
| HGB | NM | 718-7 | HGB | LN | 718-7^HGB^LN |
| MCV | NM | 787-2 | MCV | LN | 787-2^MCV^LN |
| MCH | NM | 785-6 | MCH | LN | 785-6^MCH^LN |
| MCHC | NM | 786-4 | MCHC | LN | 786-4^MCHC^LN |
| RDW_CV | NM | 788-0 | RDW-CV | LN | 788-0^RDW-CV^LN |
| RDW_SD | NM | 21000-5 | RDW-SD | LN | 21000-5^RDW-SD^LN |
| HCT | NM | 4544-3 | HCT | LN | 4544-3^HCT^LN |
| PLT | NM | 777-3 | PLT | LN | 777-3^PLT^LN |
| MPV | NM | 32623-1 | MPV | LN | 32623-1^MPV^LN |
| PDW | NM | 32207-3 | PDW | LN | 32207-3^PDW^LN |
| PCT | NM | 10002 | PCT | 99MRC | 10002^PCT^99MRC |
| RET | NM | 14196-0 | RET# | LN | 14196-0^RET#^LN |
| RET_PER | NM | 4679-7 | RET% | LN | 4679-7^RET%^LN |
| IRF | NM | 33516-6 | IRF | LN | 33516-6^IRF^LN |
| LFR | NM | 10015 | LFR | 99MRC | 10015^LFR^99MRC |
| MFR | NM | 10016 | MFR | 99MRC | 10016^MFR^99MRC |
| HFR | NM | 10017 | HFR | 99MRC | 10017^HFR^99MRC |
| NRBC | NM | 30392-5 | NRBC# | LN | 30392-5^NRBC#^LN |
| NRBC_PER | NM | 26461-4 | NRBC% | LN | 26461-4^NRBC%^LN |
| P_LCR | NM | 10014 | PLCR | 99MRC | 10014^PLCR^99MRC |
| P_LCC | NM | 10013 | PLCC | 99MRC | 10013^PLCC^99MRC |
| RBC-O | NM | 10018 | RBC-O | 99MRC | 10018^RBC-O^99MRC |
| PLT-O | NM | 10019 | PLT-O | 99MRC | 10019^PLT-O^99MRC |

| | | | | | |
|---|---|---|---|---|---|
| PLT-I | NM | 10022 | PLT-I | 99MRC | 10022^PLT-I^99MRC |
| WBC-D | NM | 10024 | WBC-D | 99MRC | 10024^WBC-D^99MRC |
| PDW_SD | NM | 10031 | PDW_SD | 99MRC | 10031^PDW-SD^99MRC |
| IPF | NM | 10041 | IPF | 99MRC | 10041 ^ IPF ^99MRC |
| IPF# | NM | 10062 | IPF# | 99MRC | 10062^ IPF# ^99MRC |
| MRV | NM | 48706-6 | MRV | LN | 48706-6 ^ MRV ^ LN |
| RHE | NM | 10043 | RHE | 99MRC | 10043 ^ RHE ^99MRC |
| Neuts Band%. Manual | NM | 764-1 | Neuts Band%. Manual | LN | 764-1 ^ Neuts Band%. Manual ^LN |
| Neuts Seg%. Manual | NM | 769-0 | Neuts Seg%. Manual | LN | 769-0 ^ Neuts Seg%. Manual ^ LN |
| Abnormal Lymphs%. Manual | NM | 29261-5 | Abnormal Lymphs%. Manual | LN | 29261-5 ^ Abnormal Lymphs%. Manual ^LN |
| Pla-Aly% | NM | 33835-0 | Pla-Aly% | 99MRC | 33835-0 ^ Pla-Aly% ^99MRC |
| Mon-Aly% | NM | 4662-3 | Mon-Aly% | 99MRC | 4662-3 ^ Mon-Aly% ^99MRC |
| Imm-Aly% | NM | 10046 | Imm-Aly% | 99MRC | 10046 ^ Imm-Aly% ^99MRC |
| Other-Aly% | NM | 10047 | Other-Aly% | 99MRC | 10047 ^ Other-Aly% ^99MRC |
| Metamyelocyte%. Manual | NM | 740-1 | Metamyelocyte%. Manual | LN | 740-1 ^ Metamyelocyte%. Manual ^LN |
| Myelocytes%. Manual | NM | 749-2 | Myelocytes%. Manual | LN | 749-2 ^ Myelocytes%. Manual ^LN |
| Promyelocytes%. Manual | NM | 783-1 | Promyelocytes%. Manual | LN | 783-1 ^ Promyelocytes%. Manual ^ LN |
| Imm-Eos% | NM | 33803-8 | Imm-Eos% | 99MRC | 33803-8 ^ Imm-Eos% ^99MRC |
| Imm-Bas% | NM | 33786-8 | Imm-Bas% | 99MRC | 33786-8 ^ Imm-Bas% ^99MRC |
| Blast% | NM | 10049 | Blast% | 99MRC | 10049 ^ Blast% ^99MRC |
| Myeloblasts%. Manual | NM | 747-6 | Myeloblasts%. Manual | LN | 747-6 ^ Myeloblasts%. Manual ^LN |
| Monoblasts%. Manual | NM | 33840-0 | Monoblasts%. Manual | LN | 33840-0 ^ Monoblasts%. Manual ^ LN |
| Lymphoblasts%. Manual | NM | 33831-9 | Lymphoblasts%. Manual | LN | 33831-9 ^Lymphoblasts%. Manual ^LN |

| | | | | | |
|---|---|---|---|---|---|
| IMG/Blast% | NM | 10048 | IMG/Blast% | 99MRC | 10048 ^ IMG/Blast% ^99MRC |
| Prolymphocytes%. Manual | NM | 6746-2 | Prolymphocytes%. Manual | LN | 6746-2 ^ Prolymphocytes%. Manual ^LN |
| Promonocytes%. Manual | NM | 13599-6 | Promonocytes%. Manual | LN | 13599-6 ^ Promonocytes%. Manual ^ LN |
| Plsm-cell% | NM | 40492-1 | Plsm-cell% | 99MRC | 40492-1 ^ Plsm-cell% ^99MRC |
| WBC_O | NM | 10051 | WBC-O | 99MRC | 10051^ WBC-O ^99MRC |
| RPI | NM | 10060 | RPI | 99MRC | 10060^ RPI ^99MRC |
| Intermediate Data of Analysis Results histogram and scatter-gram data of WBC, RBC, and PLT, etc.) | | | | | |
| RBC Histogram. Binary | ED | 15050 | RBC Histogram. Binary | 99MRC | 15050^RBC Histogram. Binary^99MRC |
| RBC Histogram. Left Line | NM | 15051 | RBC Histogram. Left Line | 99MRC | 15051^RBC Histogram. Left Line^99MRC |
| RBC Histogram. Right Line | NM | 15052 | RBC Histogram. Right Line | 99MRC | 15052^RBC Histogram. Right Line^99MRC |
| RBC Histogram. Binary Meta Length | NM | 15053 | RBC Histogram. Binary Meta Length | 99MRC | 15053^RBC Histogram. Binary Meta Length^99MRC |
| RBC Histogram. Left Line Adjusted | IS | 15054 | RBC Histogram. Left Line Adjusted | 99MRC | 15054^RBC Histogram. Left Line Adjusted^99MRC |
| RBC Histogram. Right Line Adjusted | IS | 15055 | RBC Histogram. Right Line Adjusted | 99MRC | 15055^RBC Histogram. Right Line Adjusted^99MRC |
| RBC Histogram. BMP | ED | 15056 | RBC Histogram. BMP | 99MRC | 15056^RBC Histogram. BMP^99MRC |
| RBC Histogram. Total | NM | 15057 | RBC Histogram. Total | 99MRC | 15057^RBC Histogram. Total^99MRC |

| | | | | | |
|---|---|---|---|---|---|
| PLT Histogram. Binary | ED | 15100 | PLT Histogram. Binary | 99MRC | 15100^PLT Histogram. Binary^99MRC |
| PLT Histogram. Left Line | NM | 15111 | PLT Histogram. Left Line | 99MRC | 15111^PLT Histogram. Left Line^99MRC |
| PLT Histogram. Right Line | NM | 15112 | PLT Histogram. Right Line | 99MRC | 15112^PLT Histogram. Right Line^99MRC |
| PLT Histogram. Binary Meta Length | NM | 15113 | PLT Histogram. Binary Meta Length | 99MRC | 15113^PLT Histogram. Binary Meta Length^99MRC |
| PLT Histogram. Left Line Adjusted | IS | 15114 | PLT Histogram. Left Line Adjusted | 99MRC | 15114^PLT Histogram. Left Line Adjusted^99MRC |
| PLT Histogram. Right Line Adjusted | IS | 15115 | PLT Histogram. Right Line Adjusted | 99MRC | 15115^PLT Histogram. Right Line Adjusted^99MRC |
| PLT Histogram. BMP | ED | 15116 | PLT Histogram. BMP | 99MRC | 15116^PLT Histogram. BMP^99MRC |
| PLT Histogram. Total | NM | 15117 | PLT Histogram. Total | 99MRC | 15117^PLT Histogram. Total^99MRC |
| ScattergramParaVer | NM | 15014 | Scattergram ParaVer | 99MRC | 15014^ScattergramParaVer^99MRC |
| ScattergramGraphicFlags | ED | 15015 | Scattergram GraphicFlags | 99MRC | 15015^ScattergramGraphicFlags^99MRC |
| WBC DIFF Scattergram. BMP | ED | 15200 | WBC DIFF Scattergram. BMP | 99MRC | 15200^WBC DIFF Scattergram. BMP^99MRC |
| WBC DIFF Scattergram. Meta len | NM | 15203 | WBC DIFF Scattergram. Meta len | 99MRC | 15203^WBC DIFF Scattergram. Meta len^99MRC |
| WBC DIFF Scattergram. Fsc dimension | NM | 15205 | WBC DIFF Scattergram. Fsc dimension | 99MRC | 15205^WBC DIFF Scattergram. Fsc dimension^99MRC |

| | | | | | |
|---|---|---|---|---|---|
| WBC DIFF Scattergram. Ssc dimension | NM | 15206 | WBC DIFF Scattergram. Ssc dimension | 99MRC | 15206^WBC DIFF Scattergram. Ssc dimension^99MRC |
| WBC DIFF Scattergram. FL dimension | NM | 15207 | WBC DIFF Scattergram. FL dimension | 99MRC | 15207^WBC DIFF Scattergram. FL dimension^99MRC |
| WBC DIFF Scattergram. FSC-LOG dimension | NM | 15208 | WBC DIFF Scattergram. FSC-LOG dimension | 99MRC | 15208^WBC DIFF Scattergram. FSC-LOG dimension^99MRC |
| WBC DIFF Scattergram. BIN | ED | 15201 | WBC DIFF Scattergram. BIN | 99MRC | 15201^WBC DIFF Scattergram. BIN^99MRC |
| RET Scattergram. BMP | ED | 15300 | RET Scattergram. BMP | 99MRC | 15300^RET Scattergram. BMP^99MRC |
| PLT-O Scattergram. BMP | ED | 15301 | PLT-O Scattergram. BMP | 99MRC | 15301^PLT-O Scattergram. BMP^99MRC |
| RET-EXT Scattergram. BMP | ED | 15302 | RET-EXT Scattergram. BMP | 99MRC | 15302^RET-EXT Scattergram. BMP^99MRC |
| RET Scattergram. Fsc dimension | NM | 15303 | RET Scattergram. Fsc dimension | 99MRC | 15303^RET Scattergram. Fsc dimension^99MRC |
| RET Scattergram. Ssc dimension | NM | 15304 | RET Scattergram. Ssc dimension | 99MRC | 15304^RET Scattergram. Ssc dimension^99MRC |
| RET Scattergram. FL dimension | NM | 15305 | RET Scattergram. FL dimension | 99MRC | 15305^RET Scattergram. FL dimension^99MRC |
| RET Scattergram. BIN | ED | 15306 | RET Scattergram. BIN | 99MRC | 15306^RET Scattergram. BIN^99MRC |
| RET Scattergram. Meta Len | NM | 15307 | RET Scattergram. Meta Len | 99MRC | 15307^RET Scattergram. Meta Len^99MRC |
| RET Scattergram | NM | 15308 | RET | 99MRC | 15308^RET Scattergram |

| | | | | | |
|---|---|---|---|---|---|
| FSC-LOG dimension | | | Scattergram FSC-LOG dimension | | FSC-LOG dimension^99MRC |
| Flags of Abnormal Blood Cell Differential or Morphology | | | | | |
| WBC Abnormal scattergram | IS | 12000 | WBC Abnormal scattergram | 99MRC | 12000^WBC Abnormal scattergram^99MRC |
| Leucocytosis | IS | 12002 | Leukocytosis | 99MRC | 12002^Leucocytosis^99MRC |
| Leucopenia | IS | 12003 | Leukopenia | 99MRC | 12003^Leucopenia^99MRC |
| Neutrophilia | IS | 12004 | Neutrophilia | 99MRC | 12004^Neutrophilia^99MRC |
| Neutropenia | IS | 12005 | Neutropenia | 99MRC | 12005^Neutropenia^99MRC |
| Lymphocytosis | IS | 12006 | Lymphocytosis | 99MRC | 12006^Lymphocytosis^99MRC |
| Lymphopenia | IS | 12007 | Lymphopenia | 99MRC | 12007^Lymphopenia^99MRC |
| Monocytosis | IS | 12008 | Monocytosis | 99MRC | 12008^Monocytosis^99MRC |
| Eosinophilia | IS | 12009 | Eosinophilia | 99MRC | 12009^Eosinophilia^99MRC |
| Basophilia | IS | 12010 | Basophilia | 99MRC | 12010^Basophilia^99MRC |
| Erythrocytosis | IS | 12012 | Polycythemia | 99MRC | 12012^Erythrocytosis^99MRC |
| Anemia | IS | 12014 | Anemia | 99MRC | 12014^Anemia^99MRC |
| Polycythemia | IS | 12014-1 | Polycythemia | 99MRC | 12014-1^ Polycythemia ^99MRC |
| Microcytosis | IS | 12014-2 | Microcytosis | 99MRC | 12014-2^ Microcytosis ^99MRC |
| Macrocytosis | IS | 12014-3 | Macrocytosis | 99MRC | 12014-3^ Macrocytosis ^99MRC |
| Low MCHC Alert | IS | 12014-4 | Low MCHC Alert | 99MRC | 12014-4^ Low MCHC Alert ^99MRC |
| High MCHC Alert | IS | 12014-5 | High MCHC Alert | 99MRC | 12014-5^ High MCHC Alert ^99MRC |
| Thrombocytopenia | IS | 12014-6 | Thrombocytopenia | 99MRC | 12014-6^ Thrombocytopenia ^99MRC |
| Thrombocytosis | IS | 12017 | Thrombocytosis | 99MRC | 12017^Thrombocytosis^99MRC |
| Band cell suspected? | IS | 12017-1 | Band cell suspected | 99MRC | 12017^ Band neutrophils suspected ^99MRC |
| Atypical Lymphs? | IS | 15192-8 | Atypical Lymphs? | LN | 15192-8^Atypical Lymphs?^LN |
| rstRBC | IS | 34525-6 | RBC Lyse Resistance? | LN | 34525-6^rstRBC^LN |
| Anisocytosis | IS | 15150-6 | Anisocytosis | LN | 15150-6^Anisocytosis^LN |
| Imm Granulocytes? | IS | 34165-1 | Immature Gran? | LN | 34165-1^Imm Granulocytes?^LN |

| | | | | | |
|---|---|---|---|---|---|
| Platelet Clump? | IS | 7796-6 | Platelet Clump? | LN | 7796-6^Platelet Clump?^LN |
| Sample Abnormal | IS | 12021 | Sample Abnormal | 99MRC | 12021^Sample Abnormal^99MRC |
| RBC Dual Pop | IS | 10379-6 | RBC Dual Pop | LN | 10379-6^RBC Dual Pop^LN |
| RBC-CH Error | IS | 12030 | RBC-CH Error | 99MRC | 12030^RBC-CH Error^99MRC |
| HGB Sampling Error | IS | 12032 | HGB Sampling Error | 99MRC | 12032^HGB Sampling Error^99MRC |
| PLT-CH Error | IS | 12033 | PLT-CH Error | 99MRC | 12033^PLT-CH Error^99MRC |
| BASO-CH Error | IS | 12035 | BASO-CH Error | 99MRC | 12035^BASO-CH Error^99MRC |
| RET Sampling Error | IS | 12038 | RET Sampling Error | 99MRC | 12038^RET Sampling Error^99MRC |
| RET-CH Error | IS | 12039 | RET-CH Error | 99MRC | 12039^RET-CH Error^99MRC |
| RET Abn Scattergram | IS | 12040 | RET Abn Scattergram | 99MRC | 12040^RET Abn Scattergram^99MRC |
| Reticulocytosis | IS | 12041 | Reticulocytosis | 99MRC | 12041^Reticulocytosis^99MRC |
| NRBC-CH Error | IS | 12043 | NRBC-CH Error | 99MRC | 12043^NRBC-CH Error^99MRC |
| Lipid Particles? | IS | 12055 | Lipid Particles? | 99MRC | 12055^Lipid Particles? ^99MRC |
| Aspiration Abn | IS | 12101 | Aspiration Abn | 99MRC | 12101^ Aspiration Abn ^99MRC |
| Aspiration Abnormal | IS | 12105 | Aspiration Abnormal | 99MRC | 12105^ Aspiration Abnormal ^99MRC |

## 5.11 Analysis Parameter

The analysis parameter unit is displayed as standard unit.

Table 5-3 Parameter Units in Analysis

| **Units in Software** | **Parameter Units in Communication (OBX-6)** |
|---|---|
| 10^12/L | 10*12/L |
| 10^9/L | 10*9/L |
| 10^6/uL | 10*6/uL |
| 10^4/uL | 10*4/uL |
| 10^3/uL | 10*3/uL |
| 10^2/uL | 10*2/uL |
| mL/L | mL/L |
| /nL | /nL |
| /pL | /pL |
| g/L | g/L |
| g/dL | g/dL |
| L/L | L/L |
| mmol/L | mmol/L |
| % | % |
| fL | fL |

| | |
|---|---|
| um^3 | um3 |
| pg | pg |
| fmol | fmol |
| amol | amol |
| year (age unit) | yr |
| month (age unit) | mo |
| day (age unit) | d |
| hour (age unit) | hr |
| week (age unit) | wk |

# 6 Enumeration Value of the Other HL7 Field

| Data | Value Enumeration |
|---|---|
| Take Mode | Value enumeration:<br>“O” - open-vial<br>“A” - autoloading<br>“C” – closed-tube |
| Blood Mode | Value enumeration:<br>“W”- whole blood<br>“P” - predilute<br>“Q” – control |
| Test Mode | Value enumeration:<br>Can be one of the following modes, or any combination of the modes:<br>“CBC”<br>“DIFF”<br>“RET”<br>“NRBC” |
| Species | Value enumeration:<br>“720” - Dog<br>“721” - Cat<br>“722” - Horse<br>“723” - Rat<br>“724” - Mouse<br>“725” - Rabbit<br>“726” - Monkey<br>“727” - Pig<br>“728” - Camel<br>“729” - Goat<br>“730” - Ferret<br>“731” - Cow<br>“732” - Sheep<br>“733” - Llama<br>“734” – Panda<br><br>“1000” -SelfDefine1<br>“1001” -SelfDefine2<br>…<br>“1119” -SelfDefine200 |
| Qc Level | Value enumeration:<br>“L”- Low<br>“M”- Normal<br>“H”- High |
| Histogram discriminator adjusted mark and re-exam flag and other flags | The data type of OBX-2 is "IS". Value enumeration: |

| | |
|---|---|
| | “T”- True<br>“F”- False |
| Gender | “M”/“m”: Male<br>“F”/“f”: Female<br>“U”/“u”: unknow<br>Others: displayed as strings |

1. Histogram data: the histograms can be transmitted in the following ways based on the software configuration:

1) Do not transmit histogram data.

2) Transmitted as bitmap. The data type field of OBX segment is “ED”, and the data field is in the form of “^Image^BMP^Base64^……bitmap histogram data……”, where “Image” indicates that the data in transmission is data of graphs, “BMP” is the custom subdata type, and “Base64” is the way of coding the bitmap data.

3) Transmitted as binary histogram data. The data type field of OBX segment is “ED”, and the data field is in the form of “^Application^Octet-stream^Base64^……histogram data……”, where “Application^Octer-stream” is the HL7 standard subdata type, indicating the binary data defined by the application, and “Base64” is the way of coding the bitmap data.

Note: the ID field in the OBX segment defines whether the histogram is transmitted in bitmap or binary data.

2. Scattergram data: the data type field of OBX segment is “ED”, and the data field is in the form of “^Image^BMP^Base64^……scattergram bitmap data……”, where “Image^BMP^Base64” indicates that the data in transmission is BMP data coded by Base 64.

## 6.1 HL7 Data Type Definition

- CE - Code Element

<identifier (ST)> ^ <text (ST)> ^ <name of coding system (ST)> ^ <alternate identifier (ST)> ^ <alternate text (ST)> ^ <name of alternate coding system (ST)>

- CM - Composite

Format defined by the field.

- CX - Extended composite ID with check digit

<ID (ST)> ^ <check digit (ST)> ^ <code identifying the check digit scheme employed (ID)> ^ < assigning authority (HD)> ^ <identifier type code (IS)> ^ < assigning facility (HD)>

- ED – Encapsulate Data

<source application (HD)> ^ <type of data (ID)> ^ <data sub type (ID)> ^ <encoding (ID)> ^ <data (ST)>

- EI - Entity Identifier

<entity identifier (ST)> ^ <namespace ID (IS)> ^ <universal ID (ST)> ^ <universal ID type (ID)>

- FC – Financial Class

<financial class (IS)> ^ <effective date (TS)>

- HD - Hierarchic designator

<namespace ID (IS)> ^ <universal ID (ST)> ^ <universal ID type (ID)>

Used only as part of EI and other data types.

- FT - Formatted text

This data type is derived from the string data type by allowing the addition of embedded formatting instructions. These instructions are limited to those that are intrinsic and independent of the circumstances under which the field is being used.

- IS - Coded value for user-defined tables

The value of such a field follows the formatting rules for an ST field except that it is drawn from a site-defined (or user-defined) table of legal values. There shall be an HL7 table number associated with IS data types.

- ID - Coded values for HL7 tables

The value of such a field follows the formatting rules for an ST field except that it is drawn from a table of legal values. There shall be an HL7 table number associated with ID data types.

- NM - Numeric

A number represented as a series of ASCII numeric characters consisting of an optional leading sign (+ or -), the digits and an optional decimal point.

- PL - Person location

<point of care (IS )> ^ <room (IS )> ^ <bed (IS)> ^ <facility (HD)> ^ < location status (IS )> ^ <person location type (IS)> ^ <building (IS )> ^ <floor (IS )> ^ <location description (ST)>

- PT - Processing type

<processing ID (ID)> ^ <processing mode (ID)>

- SI - Sequence ID

A non-negative integer in the form of an NM field. The uses of this data type are defined in the chapters defining the segments and messages in which it appears.

- ST – String

- TS - Time stamp

YYYY[MM[DD[HHMM[SS[.S[S[S[S]]]]]]]][+/-ZZZZ] ^ <degree of precision>

- XCN - Extended composite ID number and name

In Version 2.3, use instead of the CN data type. <ID number (ST)> ^ <family name (ST)> & <last_name_prefix (ST) ^ <given name (ST)> ^ <middle initial or name (ST)> ^ <suffix (e.g., JR or III) (ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (ST)> ^ <source table (IS)> ^ <assigning authority (HD)> ^ <name type code (ID)> ^ <identifier check digit (ST)> ^ <code identifying the

check digit scheme employed (ID)> ^ <identifier type code (IS)> ^ <assigning facility (HD)> ^ <name representation code (ID)>

- XPN - Extended person name

In Version 2.3, replaces the PN data type. <family name (ST)> ^ <given name (ST)> & <last_name_prefix (ST)> ^ <middle initial or name (ST)> ^ <suffix (e.g., JR or III) (ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (IS)> ^ <name type code (ID) > ^ <name representation code (ID)>

- VID - Version identifier

<version ID (ID)> ^ <internationalization code (CE)> ^ <international version ID (CE)>

## 6.2 Binary Data Communication

Histogram data is transmitted the binary. :The data type field of OBX segment is "ED", and the data field is in the form of ^Application^Octer-stream^Base64^……histogram data……", meanwhile the "Application" indicates that application program data is transmitted, "Octer-stream" indicates the data is of byte stream type, "Base64" indicates the coding mode of the bitmap data.

The scattergram binary data transmission is the similar as the histogram data.

Scattergram, bitmap data communication: The data type field of OBX segment is "ED", and the data field is in the form of "^Image^BMP^Base64^……scattergram bitmap data……", "Image^BMP^Base64" indicates that the data in transmission is BMP data coded by Base 64.

## 6.3 Base64 Coding

1. Select the 3 adjacent bytes (i.e. 24 bit) from the data stream to be encoded; from left to right, divide them into 4 6-bit groups; and then, the ASCII string is obtained by mapping based on Table 14 below.

| | | | | |
|---|---|---|---|---|
| Raw data:: | 15H | A3H | 4BH | |
| Binary data | 00010101 | 10100011 | 01001011 | |
| 6-bit groups obtained after dividing | 000101 | 011010 | 001101 | 001011 |
| Corresponding codes | 5H | 1AH | 0DH | 0BH |
| Corresponding characters | F | a | N | L |

Table 6-1 Base64 Mapping

| Value/Code | Value/Code | Value/Code | Value/Code |
|---|---|---|---|
| 0 A | 17 R | 34 I | 51 z |
| 1 B | 18 S | 35 j | 52 0 |
| 2 C | 19 T | 36 k | 53 1 |
| 3 D | 20 U | 37 l | 54 2 |
| 4 E | 21 V | 38 m | 55 3 |
| 5 F | 22 W | 39 n | 56 4 |
| 6 G | 23 X | 40 o | 57 5 |
| 7 H | 24 Y | 41 p | 58 6 |
| 8 I | 25 Z | 42 q | 59 7 |
| 9 J | 26 a | 43 r | 60 8 |
| 10 K | 27 b | 44 s | 61 9 |
| 11 L | 28 c | 45 t | 62 + |
| 12 M | 29 d | 46 u | 63 / |
| 13 N | 30 e | 47 v | |
| 14 O | 31 f | 48 w | (pad) = |
| 15 P | 32 g | 49 x | |
| 16 Q | 33 h | 50 y | |

2. Repeat step 1 continuously till the whole data stream is encoded.

When the data left is less than 3 bytes, 0 is added to the right to complement. If the 6-bit groups obtained is composed of the complement bit (0) only, then it is mapped to the “=“ character. When there is the last one byte left, there will be two “=“ characters in the obtained coding string; when two bytes are left, then the obtained coding string consists of one “=“ character. See the two examples below:

① Raw data 0AH

00001010

Data obtained after 00001010 00000000 00000000

6-bit groups obtained after dividing 000010 100000 000000 000000

Corresponding codes 02H 20H 00H 00H

Corresponding characters C g = =

② Raw data 0AH 0BH

00001010 00001011

Data obtained after 00001010 00001011 00000000

6-bit groups obtained after 000010 100000 101100 000000

Corresponding codes 02H 20H 2CH 00H

Corresponding characters C g s =